The Invisible Storm: Hurricane Spillovers Beyond Geographic Reach
Evidence from a Gaussian Graphical Model Approach

Pietro Fraccaroli Gimelgo Xirinda

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Abstract

This study investigates whether the economic effects of hurricanes in Florida propagate through channels beyond geographic proximity. Using monthly nighttime light intensity from NOAA VIIRS as a proxy for local economic activity and county-level FEMA Major Disaster Declarations as a measure of hurricane exposure, we construct a balanced panel of Florida's 67 counties from 2014 to 2023. The empirical strategy proceeds in two stages. First, we estimate a two-way fixed effects panel regression of nighttime light intensity on direct hurricane exposure and geographically proximate spillovers, controlling for county and month fixed effects. Second, we apply the Graphical Lasso to the regression residuals to recover a sparse Gaussian conditional dependence network across counties, and compare this learned structure with a standard geographic contiguity benchmark. The results suggest that hurricane spillovers extend well beyond shared borders. Direct hurricane exposure is associated with a statistically significant decline in nighttime light intensity, while the network estimated from the residuals reveals that most conditional dependence links are non-geographic. In particular, Orange County emerges as the dominant hub, consistent with the idea that tourism and related economic channels connect counties across the state independently of physical adjacency. An extended regression confirms that counties connected through the non-geographic glasso network experience significant declines in nighttime light intensity when their economic partners are hit by a hurricane. Taken together, the findings suggest that standard spatial matrices based solely on geographic proximity understate the true reach of hurricane-driven economic shocks in Florida.

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1 Introduction

Natural disasters can affect economic activity not only in the places that are directly hit, but also in places connected through broader economic linkages. Hurricanes may disrupt tourism flows, commuting patterns, supply chains, consumer demand, and expectations at a regional scale. A growing empirical literature documents that disasters have meaningful local economic effects, including declines in income, changes in migration, and pressure on housing markets (Boustan et al., 2020). In parallel, another literature has shown that satellite-based nighttime lights provide a useful proxy for local economic activity, particularly in settings where conventional subnational output measures are not available at high frequency (Bertinelli and Strobl, 2013; Ishizawa et al., 2017). Together, these literatures suggest that hurricane episodes offer a useful setting in which to study both direct local impacts and broader patterns of economic propagation.

At the same time, most empirical analyses of spatial spillovers rely on a pre-specified spatial structure, usually built from geographic contiguity or distance. Such choices are intuitive and often convenient, but they may fail to capture economically meaningful linkages between locations. Counties that do not share a border may nonetheless be strongly connected through tourism circuits, transport corridors, labor flows, or common demand conditions, while neighboring counties may in practice have weak economic ties. This concern is not new. The spatial-econometrics literature has long recognized that empirical results can be sensitive to the choice of the spatial weights matrix and that the relevant interaction structure is often uncertain rather than known (Bhattacharjee and Jensen-Butler, 2013). In other words, the common practice of imposing adjacency may be useful as a benchmark, but it may also be restrictive.

This project asks whether the economic effects of major hurricanes in Florida propagate through channels beyond geographic proximity. We examine county-level variation in local economic activity using monthly nighttime light intensity from the NOAA VIIRS product (**?**). Florida offers a particularly attractive setting because hurricane exposure varies sharply across counties, yet the state's economy also contains obvious channels through which shocks may travel beyond landfall areas, especially through tourism, transport, and statewide demand conditions. Our substantive interest is therefore not limited to whether neighboring counties are affected, but whether the broader pattern of county interdependence is richer than contiguity alone suggests.

The project's main contribution is both substantive and methodological. Substantively, it studies whether hurricane-induced economic disruption in Florida extends beyond directly hit or adjacent counties. Methodologically, it combines a conventional panel-data approach with a data-driven network-learning step. Rather than relying only on a fixed adjacency matrix, we use a Gaussian graphical model estimated through the Graphical Lasso to recover a sparse network of conditional dependence across Florida counties (Friedman et al., 2008). In Gaussian graphical models, non-zero off-diagonal entries in the precision matrix correspond to direct conditional dependence, whereas zero entries correspond to conditional independence after conditioning on all remaining variables (Dempster, 1972; Meinshausen and Bühlmann, 2006). This framework

allows us to ask whether the dependence structure observed in county-level economic activity resembles the geography-based benchmark or instead points to broader economic propagation channels.

This approach also connects to a growing literature that uses partial-correlation networks to study connectedness and shock transmission in economics and finance. Anufriev and Panchenko (2015), for example, show how covariance-based methods and Gaussian graphical models can be used to uncover economically meaningful network structures and study their implications for shock propagation. While our application differs in setting, the same intuition applies: once common influences and direct own-county effects are controlled for, residual dependence can still reveal a structured system of connections. In this sense, the project contributes to the broader question of whether dependence structures in applied economics should be fully imposed a priori or partially learned from the data.

2 Data and Measurement

This analysis combines monthly nighttime light data with county-level hurricane exposure across Florida's 67 counties over the period January 2014 – December 2023. The dataset is structured as a balanced panel, with county and month fixed effects included to control for time-invariant county characteristics and common temporal shocks respectively.

2.1 Dependent Variable

Nighttime Light Intensity. We measure local economic activity using monthly nighttime radiance from the VIIRS/DNB Monthly V1 composite (`NOAA/VIIRS/DNB/MONTHLY_V1/VCMSLCFG`), produced by the National Oceanic and Atmospheric Administration (NOAA). Radiance is expressed in nanowatts per square centimetre per steradian (nW/cm^2/sr). Nighttime light intensity serves as a well-established proxy for local economic activity (Henderson et al., 2012), as areas with greater commercial, industrial, and residential activity emit more artificial light at night.

For each county, we aggregate mean radiance over county boundaries defined by the US Census TIGER/2018 shapefile, using a 500-metre resolution mean reducer applied server-side on Google Earth Engine. This yields a balanced panel of 8,040 observations (67 counties $\times$ 120 months), which constitutes the dependent variable NTL_{it}.

2.2 Treatment Variable

Hurricane Exposure. We construct a binary hurricane indicator using the FEMA Disaster Declarations Summaries (Federal Emergency Management Agency, 2024). We retain Major Disaster Declarations only, which are issued during the event for immediate resource mobilisation rather than long-term reconstruction. We further restrict to records explicitly classified as hurricanes,

excluding tropical storms and other subtypes. This yields 8 Major Disaster Declarations within our sample period, summarised in Table 1.

Table 1: Florida Hurricanes in the Sample (2014–2023)

Hurricane	**Date**	**Counties Affected**
Hurricane Hermine	September 2016	26
Hurricane Matthew	October 2016	18
Hurricane Irma	September 2017	67
Hurricane Michael	October 2018	18
Hurricane Dorian	September 2019	14
Hurricane Sally	September 2020	14
Hurricane Ian	September 2022	67
Hurricane Nicole	November 2022	61

The indicator $FEMA_{it}$ equals 1 in the month a Major Disaster Declaration was issued for county i, and 0 otherwise.

2.3 Controls

County Fixed Effects. County fixed effects α_i absorb all time-invariant differences across counties, such as permanent differences in economic size, population density, and industrial composition.

Month Fixed Effects. Month fixed effects γ_t absorb common shocks affecting all Florida counties simultaneously, including seasonal patterns in nighttime radiance and aggregate statewide economic cycles. Together, the two sets of fixed effects reduce our identifying variation to within-county deviations from the common monthly trend.

3 Empirical Strategy

3.1 Stage 1: Panel Regression

To isolate the variation in economic activity not explained by direct hurricane exposure and geographic spillovers, we estimate the following two-way fixed effects panel regression:

$$NTL_{it} = \beta_1 \cdot FEMA_{it} + \beta_2 \cdot hurricane_lag_{it} + \alpha_i + \gamma_t + \varepsilon_{it} \tag{1}$$

where NTL_{it} is the mean nighttime light radiance in county i in month t, serving as a proxy for local economic activity. $FEMA_{it}$ is a binary indicator equal to 1 in the month a Major Disaster Declaration was issued for county i, and 0 otherwise, capturing direct hurricane exposure at the county-month level. $hurricane_lag_{it}$ is the number of geographically contiguous counties of i that received a Major Disaster Declaration in month t, as defined in Section 3.3, absorbing spatially proximate spillovers along the geographic network. α_i and γ_t denote county and month fixed effects, respectively. ε_{it} is the regression residual.

While β_1 and β_2 capture the average effect of direct and geographically proximate hurricane exposure on local nighttime light intensity, the residuals ε_{it} are the central object of interest. They represent the deviation of county i in month t from what the model would predict given its structural characteristics, the common monthly trend, direct hurricane exposure, and border spillovers. It is precisely this residual variation that may reflect indirect propagation channels beyond geographic proximity, and which feeds into Stage 2.

3.2 Stage 2: Graphical Lasso

Having obtained the residuals from Stage 1, we now turn to the core of our analysis: estimating the conditional dependence structure across Florida's 67 counties. The procedure unfolds in three steps.

Residual Matrix. We arrange the residuals into a matrix $E \in \mathbb{R}^{T \times N}$, where rows index months ($T = 120$) and columns index counties ($N = 67$):

$$E = \begin{pmatrix} \varepsilon_{1,1} & \cdots & \varepsilon_{1,67} \\ \vdots & \ddots & \vdots \\ \varepsilon_{120,1} & \cdots & \varepsilon_{120,67} \end{pmatrix} \tag{2}$$

Each entry ε_{it} represents the deviation of county i from its predicted nighttime light intensity in month t. The matrix therefore captures the joint idiosyncratic variation across all counties over the entire sample period.

Step 1 — Covariance Matrix. We compute the empirical covariance matrix $S \in \mathbb{R}^{N \times N}$ from the residual matrix:

$$S = \frac{1}{T-1} E^\top E \tag{3}$$

where entry S_{ij} measures how much counties i and j co-move across all 120 months. A high value of S_{ij} indicates that the two counties tend to experience similar deviations from their predicted activity simultaneously. However, high covariance between two counties may simply reflect their common connection to a third county, rather than a direct economic link. For instance, Lee and Osceola may exhibit high covariance ($S_{Lee,\ Osceola} = 0.8$) not because they are directly linked, but because both are connected to a third county driving their joint variation.

Step 2 — Precision Matrix. To disentangle direct from indirect relationships, we apply the Graphical Lasso (Friedman et al., 2008) to estimate the sparse precision matrix $\hat{\Theta} \in \mathbb{R}^{N \times N}$, defined as the penalised inverse of the covariance matrix:

$$\hat{\Theta} = \arg\max_{\Theta \succ 0} \left[\log\det\Theta - tr(S\Theta) - \rho \|\Theta\|_1 \right] \tag{4}$$

where $\rho > 0$ is the penalisation parameter controlling network sparsity. A higher ρ imposes a stricter penalty on non-zero entries, retaining only the strongest direct connections. Each off-diagonal entry $\hat{\Theta}_{ij}$ encodes the partial correlation between counties i and j after conditioning on all other counties, that is, their direct conditional dependence net of all indirect pathways. Crucially, entries set to zero by the penalisation imply conditional independence: once all other counties are accounted for, no residual co-movement remains between i and j. Returning to our example, after conditioning on all other counties, Lee and Osceola retain a precision value of $\hat{\Theta}_{Lee,\ Osceola} = 0.6$, confirming a direct economic link between the two counties.

Step 3 — Adjacency Matrix. Finally, we convert the precision matrix into a binary adjacency matrix $A \in \{0,1\}^{N \times N}$:

$$A_{ij} = \mathbf{1}\left[\hat{\Theta}_{ij} \neq 0\right], \quad i \neq j \tag{5}$$

where $A_{ij} = 1$ indicates a direct conditional dependence between counties i and j, and $A_{ij} = 0$ indicates conditional independence. In our example, $A_{Lee,\ Osceola} = 1$, confirming a direct economic connection between the two counties that is not mediated by any intermediate county.

3.3 Geographic Benchmark

To control for spatially proximate propagation and to serve as a benchmark for the Graphical Lasso network, we construct a binary contiguity matrix $W \in \{0,1\}^{67 \times 67}$ from the US Census TIGER/2018 county shapefile, using rook contiguity, that is, two counties are considered neighbours if and only if they share a physical border, with $W_{ij} = 1$ if counties i and j are contiguous and $W_{ij} = 0$ otherwise.

W serves two distinct roles in our analysis. First, it is used to construct the variable $hurricane_lag_{it}$, defined as the number of geographically contiguous counties of i that received a FEMA Major Disaster Declaration in month t:

$$hurricane_lag_{it} = \sum_{j=1}^{N} W_{ij} \cdot FEMA_{jt} \tag{6}$$

Including $hurricane_lag_{it}$ in the Stage 1 regression absorbs the geographically proximate propagation channel, ensuring that the residuals entering Stage 2 reflect idiosyncratic variation purged of both direct hurricane exposure and border spillovers. Second, W serves as the geographic benchmark in Stage 2, against which the Graphical Lasso network is compared to assess whether the estimated dependency structure extends beyond physical proximity.

3.4 Extended Regression

Finally, we estimate an extended regression to directly test whether the economic network identified by the Graphical Lasso constitutes a channel of hurricane propagation beyond geographic

contiguity:

$$NTL_{it} = \alpha_i + \gamma_t + \beta_1 FEMA_{it} + \beta_2 hurricane_lag_{it} + \beta_3 glasso_neighbor_hit_{it} + \varepsilon_{it} \tag{7}$$

This specification augments the baseline regression with $glasso_neighbor_hit_{it}$, defined for each county i in month t as the number of non-geographically contiguous counties connected to i in the Graphical Lasso network that received a FEMA Major Disaster Declaration in that same month:

$$glasso_neighbor_hit_{it} = \sum_{j=1}^{N} A^*_{ij} \cdot FEMA_{jt} \tag{8}$$

where $A^*_{ij} = A_{ij} \cdot \mathbf{1}[W_{ij} = 0]$ denotes the adjacency matrix of the Graphical Lasso network restricted to non-geographic connections only, that is, pairs of counties that are connected in the glasso network but do not share a physical border.

The coefficient β_3 is the parameter of interest. A negative and statistically significant estimate would indicate that counties experience a reduction in nighttime light intensity when their non-geographic economic partners are struck by a hurricane, even in the absence of direct damage or geographic contiguity. This constitutes a direct test of our research question.

4 Results

4.1 Summary Statistics

Table 2 reports descriptive statistics for the main variables used in the analysis across the full panel of 8,040 county-month observations.

Table 2: Summary Statistics

Variable	Source	Unit	Min	Median	Mean	Max
NTL Mean	NOAA VIIRS / GEE	nW/cm^2/sr	0.025	1.526	2.871	20.140
Hurricane Hit	FEMA OpenFEMA	Binary	0	0	0.050	1
Hurricane Lag	US Census TIGER	Count	0	0	0.212	7
Glasso Neighbor Hit	FEMA / Glasso	Count	0	0	0.090	48

Nighttime light intensity exhibits substantial cross-sectional heterogeneity across Florida's 67 counties. The distribution is right-skewed, with a median of 1.53 nW/cm^2/sr and a mean of 2.87 nW/cm^2/sr, reflecting the concentration of economic activity in a small number of highly urbanised counties. The maximum value of 20.14 nW/cm^2/sr corresponds to Orange County, the most economically active county in the state, driven respectively by the Orlando tourism cluster. At the lower end, rural counties such as Franklin and Liberty record values close to zero,

consistent with their sparse population and limited commercial activity. The three hurricane-related variables are heavily zero-inflated by construction, as hurricane declarations affect 400 counties in 120 months.

To illustrate the geographic distribution of hurricane exposure, Figures 1–2–3 map the counties affected by three representative events: Hurricane Irma (2017), which received declarations across all 67 counties; Hurricane Michael (2018), concentrated in 18 counties along the northwest panhandle; and Hurricane Matthew (2016), which affected 18 counties along the northeast Atlantic coast. Together, these three events capture the full range of spatial coverage in the sample, from statewide exposure to highly localised shocks. Figure 4 maps the average number of contiguous neighbours hit by a hurricane across the sample period, which is higher for interior counties by construction, as they share more physical borders than coastal or peninsular counties.

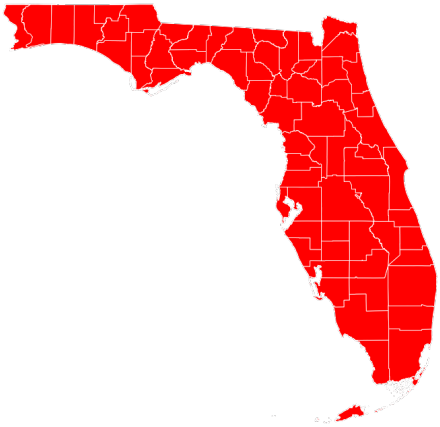

Figure 1: Counties affected by Hurricane Irma (September 2017). All 67 Florida counties received a Major Disaster Declaration.

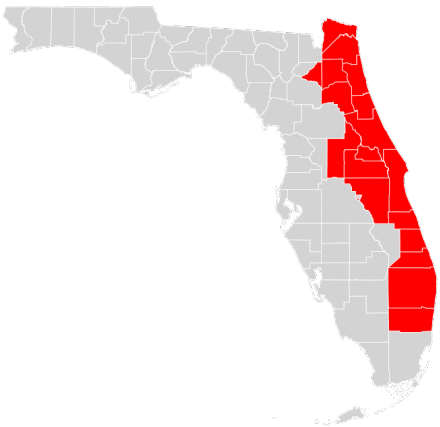

Figure 2: Counties affected by Hurricane Matthew (October 2016). Declarations were issued for 18 counties along the northeast Atlantic coast.

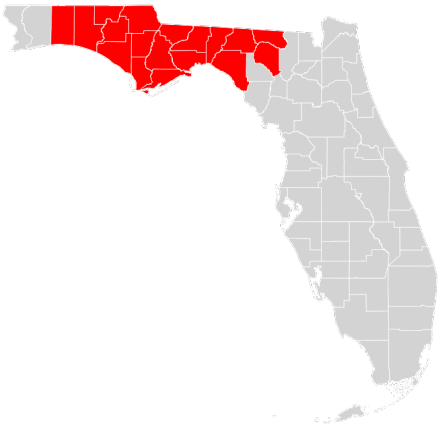

Figure 3: Counties affected by Hurricane Michael (October 2018). Declarations were issued for 18 counties concentrated in the northwest panhandle.

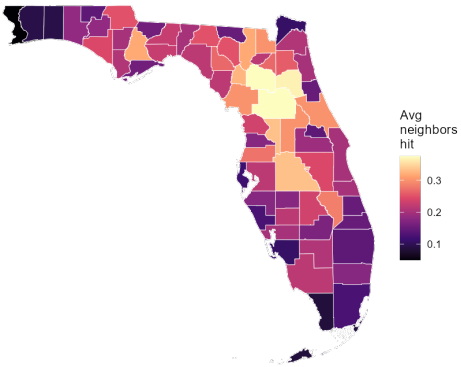

Figure 4: Average number of contiguous neighbouring counties hit by a hurricane per county (2014–2023). Higher values for interior counties reflect their greater number of shared borders relative to coastal and peninsular counties.

4.2 Stage 1: Regression Results

Table 3 reports the results of the two-way fixed effects panel regression. The coefficient on *hurricane_hit* is negative and statistically significant at a 5% significance level ($\hat{\beta}_1 = -0.091$, $p = 0.043$), confirming that a Major Disaster Declaration is associated with a reduction in nighttime light intensity of approximately 0.09 nW/cm^2/sr in the affected county in the declaration month. The coefficient on *hurricane_lag* is positive and significant at a 5% significance level ($\hat{\beta}_2 = 0.027$, $p = 0.012$), suggesting that counties whose geographic neighbours are hit experience a slight increase in nighttime light intensity. The overall R-squared is close to zero, which is expected in a two-way fixed effects specification where most variation is absorbed by the county and month fixed effects. The residuals from this regression constitute the input to Stage 2.

Table 3: Stage 1: Two-Way Fixed Effects Panel Regression

Variable	Estimate	Std. Error	t-value	p-value
Hurricane Hit	-0.0908	0.0448	-2.029	0.043 **
Hurricane Lag	0.0269	0.0107	2.516	0.012 **
County FE: Yes	Month FE: Yes	$N = 8{,}040$	$R^2 = 0.001$	
Significance codes: ** $p < 0.05$				

4.3 Stage 2: Network Structure

Figure 5 displays the non-geographic conditional dependence network estimated by the Graphical Lasso. Each line connects two Florida counties whose residual economic activity is directly conditionally dependent after controlling for all other counties, and whose connection is not explained by geographic contiguity. The network contains 66 edges in total, of which 60 are non-geographic, counties that do not share a physical border. Only 6 edges coincide with geographic contiguity. The remaining 137 geographic borders identified in W carry no corresponding conditional dependence in the glasso network.

The most striking feature of the map is the presence of a highly connected hub in central Florida, from which a large number of connections radiate both northward toward the panhandle and southward toward the peninsula. This hub corresponds to Orange County, home to the Orlando tourism cluster, suggesting that its economic activity co-moves with counties across the entire state in ways that cannot be explained by physical proximity alone.

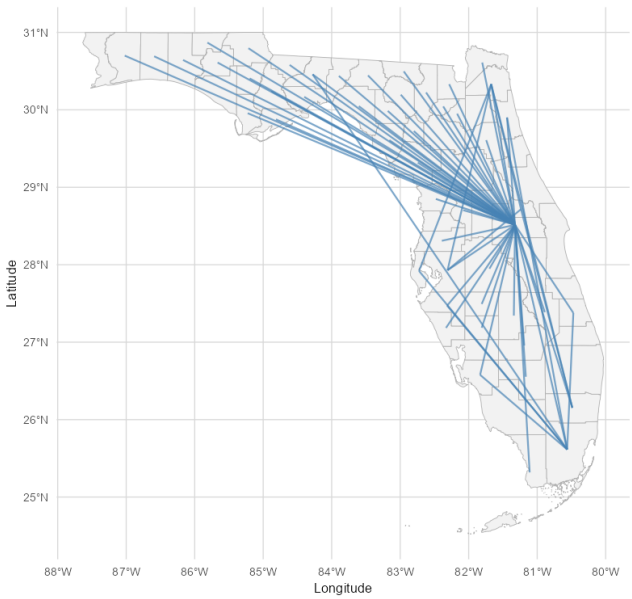

Figure 5: Non-geographic conditional dependence network estimated by the Graphical Lasso. Each line connects two counties with a non-zero partial correlation in the precision matrix $\hat{\Theta}$, restricted to pairs that are not geographically contiguous.

4.4 Extended Regression

Table 4 reports the results of the extended regression. The coefficient on *glasso_neighbor_hit* is negative, and highly significant at a 1% significance level ($\hat{\beta}_3 = -0.016$, $p < 0.001$), meaning that for each hurricane-hit county that is economically connected to county i through the glasso network, but does not share a physical border with it, county i experiences a reduction in nighttime light intensity of approximately 0.016 nW/cm^2/sr. This effect is identified after controlling for direct hurricane exposure, geographic spillovers, and the full set of county and month fixed effects. The inclusion of *glasso_neighbor_hit* also increases the F-statistic from 3.26 to 11.05 and raises the R-squared from 0.001 to 0.004, indicating a meaningful improvement in fit relative to Stage 1.

Table 4: Extended Regression: Two-Way Fixed Effects Panel Regression with Glasso Network Spillover

Variable	**Estimate**	**Std. Error**	**t-value**	**p-value**
Hurricane Hit	−0.0828	0.0447	−1.851	0.064 *
Hurricane Lag	0.0231	0.0107	2.161	0.031 **
Glasso Neighbor Hit	−0.0163	0.0032	−5.156	<0.001 ***
County FE: Yes	Month FE: Yes	$N = 8{,}040$	$R^2 = 0.004$	
Significance codes: *** $p < 0.001$, ** $p < 0.05$, * $p < 0.1$				

5 Discussion

5.1 Interpretation

The results of this analysis provide a clear answer to the research question: the economic impact of hurricanes in Florida propagates through channels that extend well beyond geographic proximity.

The Graphical Lasso network in Figure 5 reveals that 60 out of 66 estimated conditional dependence connections are non-geographic. The most striking feature of this network is the central role of Orange County, which emerges as the most connected county in the entire state. Orange County is home to Orlando and its tourism cluster, including Walt Disney World, Universal Studios, and SeaWorld, making it an economic hub whose activities are connected to counties across Florida. This suggests that tourism flows constitute a major channel through which economic shocks propagate across the state independently of physical proximity.

The extended regression in Section 4 provides a more direct and quantitative answer to the research question. The coefficient on *glasso_neighbor_hit* is negative and highly significant, indicating that counties, whose non-geographic economic partners are struck by a hurricane, experience a reduction in nighttime light intensity in the same month. Together, the network structure and the regression evidence consistently support the conclusion that hurricane impacts

travel through economic linkages, most plausibly tourism, supply chains, and labour markets, that are invisible to standard geographic proximity matrices.

5.2 Limitations

The main limitation of this analysis concerns the circularity between Stage 2 and the extended regression. The Graphical Lasso network is estimated from the residuals of Stage 1, which uses the same dependent variable, nighttime light intensity, as the extended regression in Section 4.4. In practice, this means that the variable *glasso_neighbor_hit* is constructed from a network that was itself learned from the data used in the final regression. As a result, the significance of $\hat{\beta}_3$ may be partly inflated by this dependence rather than reflecting a purely out-of-sample relationship. A more rigorous approach would estimate the Graphical Lasso on a held-out subsample, for instance, the pre-hurricane period, and use the resulting network to construct *glasso_neighbor_hit* for the full panel. We leave this refinement to future work.

A second limitation concerns the regression specification. The analysis includes only direct hurricane exposure and geographic spillovers as controls, leaving out other time-varying county-level factors that may influence nighttime light intensity, such as local labour market conditions, seasonal tourism patterns, or reconstruction spending in the aftermath of hurricanes. The inclusion of additional controls could help isolate the economic propagation channel more cleanly and reduce residual confounding.

5.3 Future Research

A natural extension of this analysis would be to expand the geographic scope from Florida to the entire United States, estimating a Graphical Lasso network across all states to identify which states are most economically connected to Florida and therefore most exposed to hurricane-driven spillovers originating in the state. More broadly, the same framework could be applied to other types of natural disasters, such as wildfires in California or floods in the Midwest, to assess whether the pattern of non-geographic economic propagation identified here generalises beyond the hurricane context.

6 Conclusion

This paper investigates whether the economic impact of hurricanes in Florida propagates through channels beyond geographic proximity, using a two-stage approach that combines panel regression with a Gaussian Graphical Model. In Stage 1, we estimate a two-way fixed effects regression of nighttime light intensity on hurricane exposure and geographic spillovers, extracting residuals that capture idiosyncratic economic variation net of direct damage. In Stage 2, we apply the Graphical Lasso to these residuals to estimate a sparse conditional dependence network across Florida's 67 counties, and compare it against a geographic contiguity benchmark.

The results show that 60 out of 66 estimated economic connections are non-geographic, with Orange County emerging as the dominant hub of the network. The extended regression confirms that counties whose non-geographic economic partners are hit by a hurricane experience a statistically significant reduction in nighttime light intensity, even in the absence of border contiguity. These findings suggest that standard spatial weight matrices based on geographic proximity substantially understate the true reach of hurricane-driven economic shocks, and that economic linkages, most plausibly through tourism and related industries, constitute an important and under explored propagation channel.

References

Anufriev, M. and Panchenko, V. (2015), 'Connecting the dots: Econometric methods for uncovering networks with an application to the australian financial institutions', *Journal of Banking & Finance* **61**, S241–S255.
URL: *https://ideas.repec.org/a/eee/jbfina/v61y2015is2ps241-s255.html*

Bertinelli, L. and Strobl, E. (2013), 'Quantifying the local economic growth impact of hurricane strikes: An analysis from outer space for the caribbean', *Journal of Applied Meteorology and Climatology* **52**(8), 1688–1697.
URL: *https://journals.ametsoc.org/view/journals/apme/52/8/jamc-d-12-0258.1.xml*

Bhattacharjee, A. and Jensen-Butler, C. (2013), 'Estimation of the spatial weights matrix under structural constraints', *Regional Science and Urban Economics* **43**(4), 617–634.
URL: *https://www.sciencedirect.com/science/article/abs/pii/S0166046213000288*

Boustan, L., Kahn, M., Rhode, P. and Yanguas, M. L. (2020), 'The effect of natural disasters on economic activity in us counties: A century of data', *Journal of Urban Economics* **118**, 103257.
URL: *https://econpapers.repec.org/RePEc:eee:juecon:v:118:y:2020:i:c:s0094119020300280*

Dempster, A. P. (1972), 'Covariance selection', *Biometrics* **28**(1), 157–175.
URL: *https://www-personal.umich.edu/ jizhu/jizhu/covar/Dempster-Biometrics72.pdf*

Federal Emergency Management Agency (2024), 'Disaster declarations summaries v2', OpenFEMA Dataset. Last accessed: May 2025.
URL: *https://www.fema.gov/openfema-data-page/disaster-declarations-summaries-v2*

Friedman, J., Hastie, T. and Tibshirani, R. (2008), 'Sparse inverse covariance estimation with the graphical lasso', *Biostatistics* **9**(3), 432–441.

Henderson, J. V., Storeygard, A. and Weil, D. N. (2012), 'Measuring economic growth from outer space', *American Economic Review* **102**(2), 994–1028.

Ishizawa, O. A., Miranda, J. J. and Zhang, H. (2017), Understanding the impact of windstorms on economic activity from night lights in central america, Policy Research Working Paper 8124, World Bank.
URL: *https://documents1.worldbank.org/curated/en/338881498577038860/pdf/WPS8124.pdf*

Meinshausen, N. and Bühlmann, P. (2006), 'High-dimensional graphs and variable selection with the lasso', *The Annals of Statistics* **34**(3), 1436–1462.
URL: *https://projecteuclid.org/journals/annals-of-statistics/volume-34/issue-3/High-dimensional-graphs-and-variable-selection-with-the-Lasso/10.1214/009053606000000281.pdf*

7 Appendix

Table 5 reports the ten most connected counties in the Graphical Lasso network, ranked by total degree. Orange County is by far the dominant hub, with 51 total connections of which 48 are non-geographic — more than six times the degree of the second most connected county, Miami-Dade. The remaining counties show moderate connectivity, with degrees ranging from 2 to 8.

Table 5: County-Level Degree in Total and Non-Geographic Networks (Top 10)

County	**Total Degree**	**Non-Geographic Degree**
Orange	51	48
Miami-Dade	8	7
Broward	5	3
Duval	5	5
Hillsborough	4	3
Seminole	4	3
Pinellas	3	2
Lee	2	2
St. Johns	2	2
Leon	2	2

Note: Non-geographic degree counts only edges where counties do not share a physical border.

Table 6 reports the ten strongest partial correlations in the non-geographic network. All top positive connections involve Orange County, consistent with its role as the dominant economic hub. The strongest link is between Orange and Hillsborough (partial correlation = 0.26), followed by Orange–Duval and Orange–Miami-Dade. Notably, two negative partial correlations appear — Pinellas–Miami-Dade and Hamilton–Orange — suggesting that economic activity in these pairs moves in opposite directions after conditioning on all other counties, potentially reflecting substitution effects in tourism or labour markets.

Table 6: Top 10 Partial Correlations in the Non-Geographic Economic Network

County (From)	**County (To)**	**Partial Correlation**	$\lvert\rho\rvert$
Orange	Hillsborough	0.2624	0.2624
Orange	Duval	0.2346	0.2346
Orange	Miami-Dade	0.2091	0.2091
Orange	Pasco	0.1901	0.1901
Orange	Lee	0.1820	0.1820
Orange	Sarasota	0.1508	0.1508
Orange	Manatee	0.1390	0.1390
Pinellas	Miami-Dade	−0.1300	0.1300
Orange	Hamilton	−0.1123	0.1123
Union	Orange	−0.1112	0.1112

Note: Restricted to non-geographically contiguous county pairs only.

Figure 6 plots the residuals from the Stage 1 fixed-effects regression for Orange County over the full sample period. The three red dashed lines mark the landfall months of Hurricanes Irma (September 2017), Michael (October 2018), and Ian (September 2022). Each hurricane corresponds to a visible negative spike in the residuals, consistent with an economic contraction in Orange County beyond what the model would predict from direct exposure and common monthly trends alone — despite Orange County not being in the direct path of any of these storms (see Figure 5 for Orange County's position in the non-geographic network).

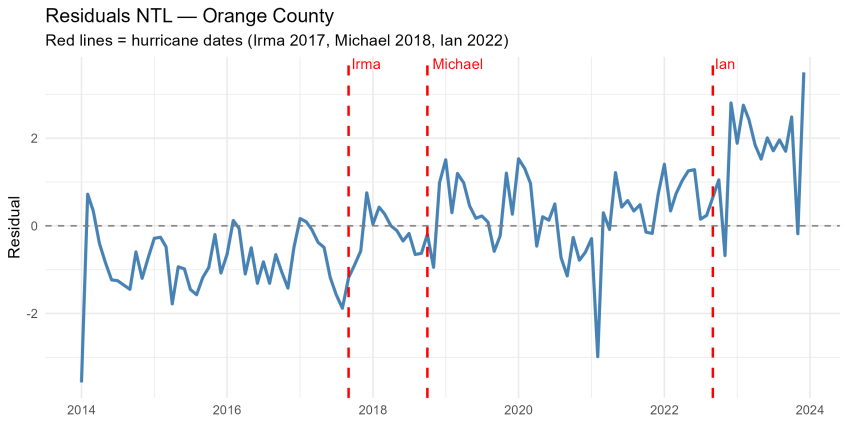

Figure 6: Residuals from the Stage 1 fixed-effects model for Orange County, 2014–2023. Red dashed lines mark the landfall months of Hurricanes Irma (September 2017), Michael (October 2018), and Ian (September 2022).